an inert gas. When no magnetic field is present, the reed blades remain separated, and the switch is in an open state, which means it does not conduct electric current. However, when a magnetic field is applied (in the present case using a fridge magnet) near the reed switch, the magnetic force causes the reed blades to attract and touch each other, closing the switch and allowing electric current to flow.

Circuit and working

Fig. 3 shows the circuit diagram of the two-way operated light cum

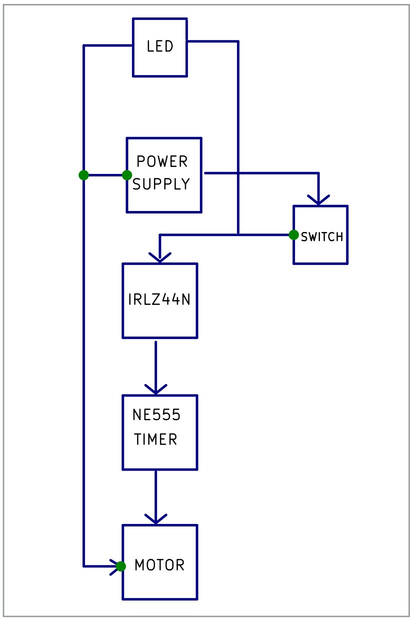

Fig. 2: Block diagram

room freshener module. It comprises NE555 timer, MOSFET IRLZ44N (T1), reed switch (S1), SPDT switch (S2), DC motor (M1), and a few other components.

Upon activating either of the switching techniques, the room freshener gets dispensed once, coinciding with a single rotation of the motor. Let's examine the step-by-step process.

The motor is interfaced with the power supply using a MOSFET and an NE555 timer IC. The switch (S2) and reed switch (S1) are connected in parallel so that either of them in a high state will activate the circuit, as one common terminal of the switches (in parallel) is linked to the negative terminal of the power supply and the other is grounded.

With an RC network consisting of resistor R3 (47k) and capacitor C2 (10µF), a trigger is generated at pin 2 of the NE555 timer IC. Pin 5 (control voltage) through capacitor C1 (1nF) is grounded. Pin 1 (ground) of the IC 555 is also connected to ground. Pin 4 (reset) and pin 8 (VCC) of IC 555 are both connected to the positive of the supply. Pin 6 (threshold) and pin 7 (discharge) of IC 555 are both connected to resistor R1 (10k) and capacitor C3 (200µF), deciding the duration of the monopulse generated (given by T = 1.1 x R1 x C3) as the IC 555 works in the monostable mode of operation. The pin 3 (output) of the NE555 timer IC is connected to the gate of MOSFET IRLZ44N via resistor R2 (10k). The source pin of IRLZ44N is connected to the ground, and its drain pin is connected to the negative terminal of motor M1. The positive of the motor is connected to the positive of the supply. The motor has capacitor C4 (100µF) wired across it to reduce radio frequency EMI caused by arcing between the brushes and commutator.

Upon activating either of the switching techniques, the room freshener is dispensed and the light switches on.

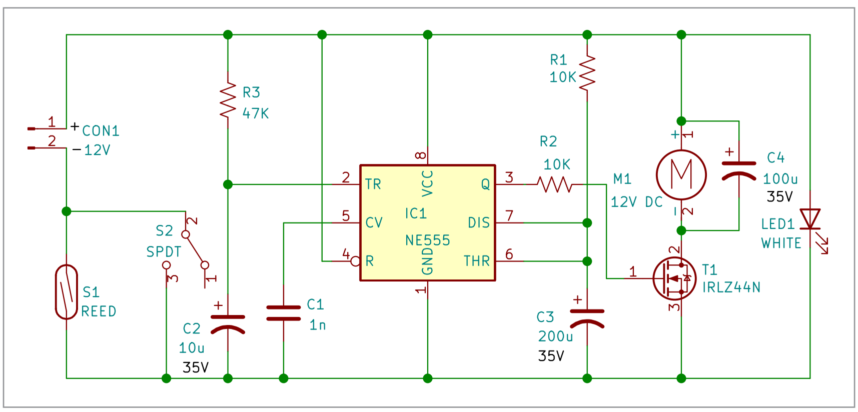

Fig. 3: Circuit diagram of the project

Construction and testing

Assemble the module on a general-purpose PCB/breadboard as per the circuit diagram. It would be helpful to refer to the author's prototype shown in Fig. 1 before assembling. After proper assembly, the module is ready to use.

For testing, bring a magnet near the reed switch (S1). When the circuit is enabled, a spray and the light would be initiated for a few seconds. In the second choice, where a switch (S2) is employed, the spray stops after a few seconds, but the light continues to glow.

The LED1 light is integrated into the system to indicate the circuit's state and to provide a source of light in case the system is kept in darkness. In conclusion, the module can prove a wonderful, personalised, low-cost alternative to the existing choices available in the market. EFY

Dr Geetali Saha is a faculty member of the Department of Electronics and Communication Engineering, GCET, Anand, Gujarat, and Dhruv Vachhani from the 3rd Year IoT Engineering Department has successfully implemented this device